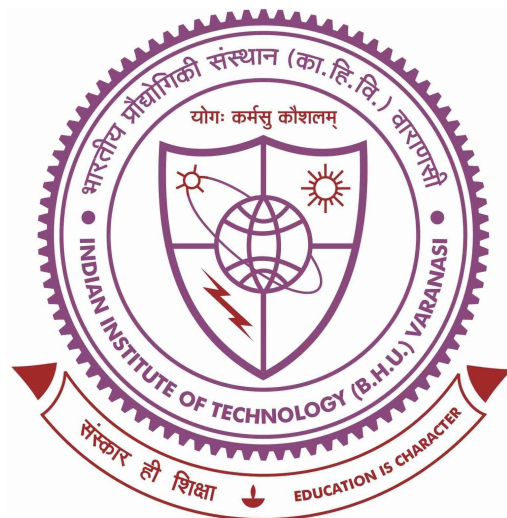


Towards phase-field modelling of phase inversion



Thesis submitted in partial fulfilment

for the Award of

INTEGRATED DUAL DEGREE (B.TECH.+ M.TECH.)

in

INDUSTRIAL CHEMISTRY

by

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Preface

This thesis presents a computational study on precipitate growth and microstructural evolution using phase-field modeling techniques. The work mainly focuses on understanding the effects of elastic interactions and interface curvature on precipitate growth through coupled Allen–Cahn and Cahn–Hilliard equations.

Phase transformations and microstructure evolution play an important role in determining the mechanical and physical properties of materials. In recent years, computational methods have become powerful tools for studying such phenomena. The present work aims to investigate these effects using numerical simulations and elastic energy formulations within the phase-field framework.

The research work presented in this thesis was carried out during my Integrated Dual Degree (B.Tech.+M.Tech.) program in the Department of Metallurgical Engineering. The study provided me with valuable experience in computational modeling, numerical methods, and scientific research.

I hope that the work presented in this thesis contributes toward a better understanding of elastic effects in precipitate evolution and phase transformations.

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Chapter 1

Introduction

1.1 Introduction

Microstructural evolution plays a crucial role in determining the mechanical, thermal, and physical properties of engineering materials. Processes such as phase separation, precipitate growth, coarsening, and solid state transformations strongly influence material behavior in alloys, ceramics, semiconductors, and functional materials. Understanding these processes is therefore important for the design and optimization of advanced materials systems.

During solid state phase transformations, the evolution of precipitates within a matrix phase is governed by thermodynamic and kinetic factors. In many alloy systems, precipitate growth occurs through diffusion-controlled mechanisms, where atoms diffuse from a supersaturated matrix into the precipitate phase. The morphology, growth rate, and spatial distribution of precipitates significantly affect material properties such as hardness, strength, creep resistance, conductivity, and phase stability.

In addition to chemical free energy, elastic interactions also play an important role in precipitate evolution. When the lattice parameters of the precipitate and matrix phases differ, elastic strain energy develops due to lattice mismatch. This misfit strain alters the local stress field and influences precipitate growth kinetics, morphology evolution, and

microstructural stability.

Traditional sharp-interface approaches, such as the Zener–Frank theory and the Laraia–Johnson–Voorhees (LJV) theory, have been widely used to study diffusion-controlled precipitate growth. However, these approaches require explicit interface tracking and become difficult to apply for complex evolving morphologies. To overcome these limitations, diffuse-interface approaches such as the phase-field method have become increasingly important.

1.2 Phase-Field Method

The phase-field method is a powerful computational technique used to model microstructural evolution without explicitly tracking interfaces. In this method, interfaces are represented as diffuse regions across which field variables vary continuously. The microstructure is described using continuous order parameters and composition fields.

The major advantage of the phase-field approach is its ability to naturally handle:

- complex interface evolution,
- precipitate coarsening,
- elastic interactions,
- morphology changes,
- and multiphase systems.

The phase-field framework is thermodynamically consistent and is based on the minimization of the total free energy of the system.

For systems involving conserved composition fields, the evolution is governed by the Cahn–Hilliard equation:

$$\frac{\partial c}{\partial t} = \nabla \cdot (M \nabla \mu) \quad (1.1)$$

where:

- c is the composition field,
- M is the mobility,
- μ is the chemical potential.

For non-conserved order parameters, the evolution is governed by the Allen–Cahn equation:

$$\frac{\partial \eta}{\partial t} = -L \frac{\delta F}{\delta \eta} \quad (1.2)$$

where:

- η is the order parameter,
- L is the relaxation coefficient,
- F is the total free energy.

The coupled solution of these equations enables simulation of precipitation, phase separation, and interface evolution.

1.3 Elastic Effects in Phase Transformations

Elastic stresses arise during phase transformations because of lattice mismatch between the matrix and precipitate phases. These stresses generate elastic strain energy, which contributes to the total free energy of the system.

The elastic contribution becomes especially important in coherent precipitates, where lattice continuity exists across the interface. The resulting elastic interactions can:

- modify precipitate growth rates,
- influence particle morphology,
- cause directional coarsening,
- and induce rafting phenomena.

The elastic energy is generally expressed as:

$$F^{el} = \frac{1}{2} \int_{\Omega} \sigma_{ij}^{el} \epsilon_{ij}^{el} d\Omega \quad (1.3)$$

where:

- σ_{ij}^{el} is the elastic stress tensor,
- ϵ_{ij}^{el} is the elastic strain tensor.

The elastic strain is obtained from:

$$\epsilon_{ij}^{el} = \epsilon_{ij} - \epsilon_{ij}^0 \quad (1.4)$$

where:

- ϵ_{ij} is the total strain,
- ϵ_{ij}^0 is the eigenstrain or misfit strain.

Mechanical equilibrium in the system is governed by:

$$\frac{\partial \sigma_{ij}}{\partial x_j} = 0 \quad (1.5)$$

Accurate computation of elastic fields is therefore essential for realistic microstructural simulations.

1.4 Numerical Methods for Elastic Field Calculation

Several numerical techniques have been developed for solving elastic equilibrium equations in phase-field models. Fourier spectral methods based on Fast Fourier Transform (FFT) are widely used because of their computational efficiency in periodic systems. These methods transform differential equations into algebraic equations in Fourier space and are particularly effective for periodic boundary conditions.

However, FFT-based spectral methods have limitations when non-periodic boundary conditions are required. Many real physical systems involve fixed, open, or non-periodic boundaries where spectral periodicity assumptions are not valid.

To address this limitation, the present work employs the Finite Difference Method (FDM) for solving the governing equations. In the FDM approach, spatial derivatives are approximated using finite difference approximations on discrete grids. This method provides flexibility in handling non-periodic boundary conditions and allows direct implementation of physically realistic boundary constraints.

Unlike FFT-based approaches, the present work solves the elastic equilibrium equations using iterative finite-difference discretization techniques under non-periodic boundary conditions.

1.5 Motivation of the Present Work

Most previous phase-field studies of precipitate growth involving elastic interactions have primarily used Fourier spectral methods with periodic boundary conditions. Although computationally efficient, these approaches may introduce artificial periodic interactions and may not accurately represent finite systems or systems with realistic physical boundaries.

The present work is motivated by the need to study precipitate growth using:

- non-periodic boundary conditions,
- finite difference discretization,
- and direct numerical solution of elastic equilibrium equations.

The implementation of finite difference techniques enables greater flexibility in modeling systems where periodicity assumptions are not appropriate.

In addition, the coupled solution of Allen–Cahn and Cahn–Hilliard equations with elastic energy contributions provides a comprehensive framework for studying diffusion-controlled

precipitate evolution.

1.6 Objectives of the Present Work

The main objectives of the present thesis are:

1. To develop a phase-field model for precipitate growth in elastically stressed systems.
2. To couple the Allen–Cahn and Cahn–Hilliard equations for microstructural evolution.
3. To incorporate elastic strain energy arising from lattice mismatch between precipitate and matrix phases.
4. To solve the elastic equilibrium equations using the Finite Difference Method (FDM).
5. To implement non-periodic boundary conditions in the numerical simulations.
6. To investigate the effect of elastic interactions on precipitate growth and morphology evolution.
7. To compare the simulated microstructural evolution with classical theoretical predictions and previous computational studies.

Chapter 2

Literature Review

2.1 Introduction

Microstructural evolution in multiphase systems has been extensively studied because of its influence on material properties. Earlier studies mainly focused on precipitate growth, coarsening, and elastic interactions during phase transformations. With the development of computational techniques, phase-field modelling became an important tool for studying such phenomena.

2.2 Classical Theory of Precipitate Growth

Zener and Frank developed the classical diffusion-controlled theory for precipitate growth in supersaturated alloys. Their model established the parabolic growth behavior of precipitates under diffusion-controlled conditions.

Later, Laraia, Johnson, and Voorhees incorporated coherent elastic strain energy into precipitate growth theory. Their work showed that elastic strain energy modifies precipitate growth kinetics and equilibrium conditions.

Although these theories provided important understanding of precipitate growth, they were based on sharp-interface assumptions and became difficult to apply for complex

evolving microstructures.

2.3 Development of Phase-Field Modelling

The phase-field method was developed to study microstructural evolution without explicit interface tracking.

Cahn and Hilliard introduced a diffuse-interface model for conserved composition evolution through the Cahn–Hilliard equation:

$$\frac{\partial c}{\partial t} = \nabla \cdot (M \nabla \mu) \quad (2.1)$$

Allen and Cahn later introduced an evolution equation for non-conserved order parameters:

$$\frac{\partial \eta}{\partial t} = -L \frac{\delta F}{\delta \eta} \quad (2.2)$$

These equations became the foundation of modern phase-field modelling and have been widely used for studying:

- phase separation,
- precipitate growth,
- coarsening,
- and morphology evolution.

2.4 Elastic Effects in Phase-Field Models

Khachaturyan developed the microelasticity theory for incorporating elastic interactions into heterogeneous systems. This framework enabled the inclusion of elastic strain energy in phase-field simulations.

Subsequent studies demonstrated that elastic interactions strongly influence:

- precipitate morphology,
- particle alignment,
- directional coarsening,
- and microstructural stability.

Elastic energy in coherent systems is generally expressed as:

$$F^{el} = \frac{1}{2} \int_{\Omega} \sigma_{ij}^{el} \epsilon_{ij}^{el} d\Omega \quad (2.3)$$

The associated mechanical equilibrium condition is:

$$\frac{\partial \sigma_{ij}}{\partial x_j} = 0 \quad (2.4)$$

2.5 Rafting and Elastic Inhomogeneity

Several studies investigated rafting behavior in elastically stressed systems.

Schmidt and Gross studied the effect of elastic modulus mismatch and applied stress on precipitate rafting. Their work showed that the direction of rafting depends on:

- **Applied stress,**
- **Eigenstrain,**
- and **Elastic Inhomogeneity.**

Gururajan and Abinandanan developed a phase-field model for studying rafting in elastically inhomogeneous systems. Their simulations demonstrated that elastic modulus mismatch significantly affects precipitate morphology and directional coarsening.

These studies suggested that elastic interactions may also influence topology evolution and phase connectivity.

2.6 Numerical Methods

Different numerical approaches have been used in phase-field simulations, including:

- **Finite Difference Method (FDM),**
- **Finite Element Method (FEM),**
- **Fourier spectral methods.**

FFT-based spectral methods are widely used because of their computational efficiency and spectral accuracy in periodic systems. However, these methods inherently assume periodic boundary conditions.

Finite Difference Method provides greater flexibility for:

- **non-periodic boundary conditions,**
- **finite-domain simulations,**
- **Iterative numerical implementation.**

Because of these advantages, FDM is suitable for studying topology evolution and connectivity changes in finite systems.

2.7 Research Gap

Most previous studies focused primarily on:

- precipitate growth,
- elastic coarsening,
- and rafting phenomena.

Relatively limited work has been carried out on phase inversion using phase-field methods with elastic interactions.

Furthermore, most elastic phase-field studies employed FFT-based spectral techniques

together with periodic boundary conditions.

The present work attempts to study topology evolution towards phase inversion using:

- coupled Allen–Cahn and Cahn-Hilliard equations,
- elastic free energy formulation,
- Finite Difference Method,
- and non-periodic boundary conditions.

Chapter 3

Methodology

3.1 Introduction

The microstructural evolution in our system is governed by the Cahn–Hilliard equation [3.1] and the Allen–Cahn equation [3.2]:

$$\frac{\partial c}{\partial t} = \nabla \cdot (M \nabla \mu) \quad (3.1)$$

$$\frac{\partial \eta}{\partial t} = -L \frac{\delta F}{\delta \eta}, \quad (3.2)$$

where M is the atomic mobility, μ is the chemical potential, and L is the relaxation coefficient for the order parameter.

Chemical potential μ is defined as the variational derivative of the total free energy per atom, F , with respect to the local composition c ,

$$\mu = \frac{\delta (F/N_v)}{\delta c}. \quad (3.3)$$

The total free energy of the system F is assumed to be given by the sum of a chemical

contribution F^{ch} and an elastic contribution F^{el} :

$$F = F^{ch} + F^{el}. \quad (3.4)$$

The chemical contribution F^{ch} is given by the following functional:

$$F^{ch} = N_v \int_{\Omega} [f(c, \eta) + K_c(\nabla c)^2 + K_{\eta}(\nabla \eta)^2] d\Omega, \quad (3.5)$$

where $f(c, \eta)$ is the bulk free energy per atom, K_c and K_{η} are the gradient energy coefficients for gradients in composition and order parameter, respectively, and NV is the number of atoms per unit volume.

The bulk free energy density $f(c, \eta)$ is given by,

$$f(c, \eta) = f^m(c) (1 - W(\eta)) + f^p(c)W(\eta) + P\eta^2(1 - \eta)^2, \quad (3.6)$$

where P , a constant, sets the height of the free energy barrier between the m-phase and p-phase.

In Eq. (3.6), $f^m(c)$ and $f^p(c)$ stand for free energies of m (matrix) and p (precipitate) phases, respectively. In our work, they are assumed to take the following simple forms:

$$f^m(c) = Ac^2, \quad (3.7)$$

$$f^p(c) = B(1 - c)^2, \quad (3.8)$$

with positive constants A and B . In Eq. (3.6), $W(\eta)$ is an interpolation function [15], given by:

$$W(\eta) = \begin{cases} 0, & \text{for } \eta < 0, \\ \eta^3(10 - 15\eta + 6\eta^2), & \text{for } 0 \leq \eta \leq 1, \\ 1, & \text{for } \eta > 1. \end{cases} \quad (3.9)$$

The elastic contribution to the free energy is:

$$F^{el} = \frac{1}{2} \int_{\Omega} \sigma_{ij}^{el} \epsilon_{ij}^{el} d\Omega, \quad (3.10)$$

where

$$\epsilon_{ij}^{el} = \epsilon_{ij} - \epsilon_{ij}^0, \quad (3.11)$$

where σ_{ij}^{el} and ϵ_{ij}^{el} are elastic stress and strain tensors, respectively, and ϵ^0 is the position-dependent eigenstrain (misfit strain). The total strain ϵ_{ij} is:

$$\epsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial r_j} + \frac{\partial u_j}{\partial r_i} \right), \quad (3.12)$$

where u_i is the displacement field.

Assuming both the m and p phases to be linearly elastic, we have:

$$\sigma_{kl}^{el} = C_{ijkl} \epsilon_{ij}^{el}, \quad (3.13)$$

where C_{ijkl} is the elastic modulus tensor.

The stress field σ_{ij}^{el} obeys the equation of mechanical equilibrium:

$$\sigma_{ij,j}^{el} = 0. \quad (3.14)$$

The eigenstrain ϵ_{ij}^0 and the elastic moduli C_{ijkl} are assumed to depend on the order parameter as follows:

$$\epsilon_{ij}^0(\eta) = \beta(\eta) e_T \delta_{ij}, \quad (3.15)$$

$$C_{ijkl}(\eta) = C_{ijkl}^{eff} + \phi(\eta) \Delta C_{ijkl}, \quad (3.16)$$

where e_T is a constant that determines the strength of the eigenstrain, δ_{ij} is the Kronecker delta, $\beta(\eta)$ and $\phi(\eta)$ are scalar interpolation functions, C_{ijkl}^{eff} is an “effective” modulus, C_{ijkl}^p and C_{ijkl}^m are the elastic moduli tensor of the p and m phases, respectively, and

$\Delta C_{ijkl} = C_{ijkl}^p - C_{ijkl}^m$. All the model parameters used in our simulations are listed in their non-dimensional form in Table 1.

Table 3.1: Parameters used in the simulations.

Parameter type	Parameter	Non-dimensional values
Elastic parameters	G^m	800
	ν	0.3
	δ	0.5, 1.0 and 2.0
	e_T	0.01
	$\phi(\eta)$	$\eta^3(10 - 15\eta + 6\eta^2) - 1/2$
	$\beta(\eta)$	$\eta^3(10 - 15\eta + 6\eta^2)$
	C_{ijkl}^{eff}	$\frac{1}{2}(C_{ijkl}^m + C_{ijkl}^p)$
Simulation parameters	Δx	0.4
	Δy	0.4
	Δt	0.2
	System size	128×128 in 2D, 512×512 in 2D, 1024×1024 in 2D
	Allowed error	1×10^{-4} in displacements

3.2 Cahn Hilliard equation

The Cahn-Hilliard equation from Eq. (3.1) can be rewritten as:

$$\frac{\partial c}{\partial t} = \nabla \cdot (M \nabla \mu)$$

where μ from Eq. (3.3) is given by:

$$\mu = \frac{\delta(F/N_v)}{\delta c}.$$

From Eq. (3.4), we have:

$$F = F^{ch} + F^{el}.$$

so, we can write μ as:

$$\mu = \frac{\delta(F^{ch}/N_v)}{\delta c} + \frac{\delta(F^{el}/N_v)}{\delta c}.$$

We get μ^{ch} by taking the functional derivative of F^{ch} , which is given by Eq. (3.5). First we substitute Eq. (3.7) and Eq. (3.8) in Eq. (3.6) and then into the Eq. (3.5) expression for

μ^{ch} , we get:

$$\mu^{ch} = 2Ac(1 - W(\eta)) + 2B(1 - c)W(\eta) - 2K_c \nabla^2.$$

The elastic contribution to the total free energy is given by Eq. (3.10):

$$F^{el} = \frac{1}{2} \int_{\Omega} \sigma_{ij}^{el} \epsilon_{ij}^{el} d\Omega.$$

To solve the Elastic contribution we have to first solve the Eq. (3.11) and Eq. (3.13) for the elastic stress(σ_{ij}^{el}) and strain tensors(ϵ_{ij}^{el}), which are given by:

$$\epsilon_{ij}^{el} = \epsilon_{ij} - \epsilon_{ij}^0,$$

$$\sigma_{kl}^{el} = C_{ijkl} \epsilon_{ij}^{el},$$

For Elastic Strain we have Eq. (3.12) and it suggests that we can get the total strain ϵ_{ij} by solving for the displacement field u_i using the following equation:

$$\epsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial r_j} + \frac{\partial u_j}{\partial r_i} \right).$$

Also, we have ϵ_{ij}^0 from Eq. (3.15) and C_{ijkl} from Eq. (3.16). After substituting these into the expression for σ_{ij}^{el} , we can then solve for the elastic stress field by solving the equation of mechanical equilibrium given by Eq. (3.14):

$$\epsilon_{ij}^0(\eta) = \beta(\eta) e_T \delta_{ij},$$

$$C_{ijkl}(\eta) = C_{ijkl}^{eff} + \phi(\eta) \Delta C_{ijkl},$$

$$\sigma_{ij,j}^{el} = 0.$$

After solving for σ_{ij}^{el} and ϵ_{ij}^{el} , we can substitute them into the expression for F^{el} and then take the functional derivative to get μ^{el} , which is the elastic contribution to the chemical potential. Finally, we can substitute μ^{ch} and μ^{el} into the Cahn-Hilliard equation to get the final form of the Cahn-Hilliard equation with elasticity. The final form of the Cahn-

Hilliard equation with elasticity is given by:

$$\frac{\partial c}{\partial t} = \nabla \cdot (M \nabla (\mu^{ch} + \mu^{el})).$$

3.3 Allen-Cahn equation

The Allen–Cahn equation from Eq. (3.2) can be written as:

$$\frac{\partial \eta}{\partial t} = -L \frac{\delta F}{\delta \eta}.$$

Using $F = F^{ch} + F^{el}$ we write

$$\frac{\delta F}{\delta \eta} = \frac{\delta F^{ch}}{\delta \eta} + \frac{\delta F^{el}}{\delta \eta}.$$

In the absence of elasticity $\delta F^{el}/\delta \eta = 0$, hence

$$\frac{\delta(F^{ch}/N_v)}{\delta \eta} = \frac{\partial f(c, \eta)}{\partial \eta} - 2K_\eta \nabla^2 \eta.$$

Using the bulk free energy density from Eq. (3.6) we obtain the explicit chemical contribution

$$\frac{\delta(F^{ch}/N_v)}{\delta \eta} = (f^p(c) - f^m(c))W'(\eta) + 2P\eta(1 - \eta)(1 - 2\eta) - 2K_\eta \nabla^2 \eta,$$

where $W'(\eta) = dW/d\eta$. Substituting $f^m(c) = Ac^2$ and $f^p(c) = B(1 - c)^2$ gives the commonly used form:

$$\frac{\partial \eta}{\partial t} = -L \left[(B(1 - c)^2 - Ac^2)W'(\eta) + 2P\eta(1 - \eta)(1 - 2\eta) - 2K_\eta \nabla^2 \eta \right] + \frac{\delta F^{el}}{\delta \eta}.$$

For the Allen–Cahn equation, the elastic contribution $\delta F^{el}/\delta \eta$ is solved in the same way as described in the previous section for the Cahn–Hilliard equation. After solving for $\delta F^{el}/\delta \eta$, we can substitute it into the above equation to get the final form of the Allen–

Cahn equation with elasticity.

3.3.1 Solving Elasticity

The elastic moduli and the eigenstrains are non-periodic on Ω (since they depend on the composition field which is non-periodic on Ω); therefore, the solution to the equilibrium equation (Eq. (3.14)) should be such that it gives rise to a strain field $e(\mathbf{r})$ that is non-periodic on Ω (up to an additive constant). The displacement field $\mathbf{u}(\mathbf{r})$ which gives rise to such non-periodic strain fields can always be written as follows [30]:

$$\mathbf{u} = \mathbf{E} \mathbf{r} + \mathbf{u}^* \quad (3.17)$$

where $\mathbf{u}^*$ is a displacement field that is non-periodic on Ω and $\mathbf{E}$ is a constant, homogeneous strain tensor (assumed to be symmetric without loss of generality, since the anti-symmetric part corresponds to a rigid rotation of the cell). It can be shown [30] that $\mathbf{E}$ denotes the mean strain tensor of the cell:

$$\langle \{e_{ij}\} \rangle = E_{ij} \quad (3.18)$$

with the following definition for the mean $\langle \{\cdot\} \rangle$ of a quantity $\{\cdot\}$:

$$\langle \{\cdot\} \rangle = \frac{1}{V} \int_{\Omega} \{\cdot\} d\Omega \quad (3.19)$$

where V is the volume of the representative domain Ω . If we denote the periodic strain by ϵ^* (such that $\langle \epsilon^* \rangle = 0$), the strain that we derive from the displacement decomposition (Eq. (3.17)) becomes

$$\epsilon_{ij} = E_{ij} + \epsilon_{ij}^* \quad (3.20)$$

where

$$\epsilon_{ij}^* = \frac{1}{2} \left(\frac{\partial u_i^*}{\partial r_j} + \frac{\partial u_j^*}{\partial r_i} \right). \quad (3.21)$$

Analogous to the expression Eq. (3.18) for the homogeneous strain, we can define the mean stress on the domain Ω as follows:

$$\langle \{\sigma_{ij}^{el}\} \rangle = \frac{1}{V} \int_{\Omega} \sigma_{ij}^{el} d\Omega \quad (3.22)$$

The mean stress thus calculated should equal the applied stress σ^A [30,31,37] [As a brief aside, we note that while Refs.[30,31] used the homogenization assumption, Jin et al.[37] used a variational approach to arrive at the above conclusion, viz., that the mean stress should equal the applied stress]. Substituting for σ_{ij}^{el} from Eq. (3.13), we obtain

$$\sigma_{ij}^A = \frac{1}{V} \int_{\Omega} C_{ijkl} (E_{kl} + \epsilon_{kl}^* - \epsilon_{kl}^0) d\Omega \quad (3.23)$$

Further, using the definition in Eq. (3.19), we can define the following mean quantities:

$$S_{ijkl} = \langle \{C_{ijkl}\} \rangle^{-1}, \quad \langle \{\sigma_{ij}^*\} \rangle = \langle \{C_{ijkl} \epsilon_{kl}^*\} \rangle, \quad \langle \{\sigma_{ij}^0\} \rangle = \langle \{C_{ijkl} \epsilon_{kl}^0\} \rangle \quad (3.24)$$

Using the Eq. (3.23) and Eq. (3.24) we obtain

$$E_{ij} = S_{ijkl} [\sigma_{kl}^A + \langle \{\sigma_{kl}^0\} \rangle - \langle \{\sigma_{kl}^*\} \rangle] \quad (3.25)$$

We may restate the elasticity problem as follows: given a non-periodic composition field c on Ω , solve the equation of mechanical equilibrium

$$\frac{\partial}{\partial r_j} [C_{ijkl} (E_{kl} + e_{kl}^* - e_{kl}^0)] = 0 \quad \text{on } \Omega \quad (3.26)$$

with the constraint

$$E_{ij} = S_{ijkl} (\sigma_{kl}^A + \langle \{\sigma_{kl}^0\} \rangle - \langle \{\sigma_{kl}^* \} \rangle) \quad (3.27)$$

and the boundary condition

$$e_{kl}^* \text{ is non-periodic on } \Omega. \quad (3.28)$$

Substituting for C_{ijkl} and e_{kl}^0 in terms of composition, and e_{kl}^* in terms of the displacement field in the equilibrium equation Eq. (3.26), and using the symmetry properties of the elastic constants and strains, we obtain

$$\begin{aligned} \frac{\partial}{\partial r_j} \left[\left(C_{ijkl}^{eff} + \phi(c) \Delta C_{ijkl} \right) \left(E_{kl} + \frac{1}{2} \left(\frac{\partial u_k^*}{\partial r_l} + \frac{\partial u_l^*}{\partial r_k} \right) - \beta(c) \epsilon_T \delta_{kl} \right) \right] &= 0. \quad (3.29) \\ \left[C_{ijkl}^{eff} \frac{\partial^2}{\partial r_j \partial r_k} + \Delta C_{ijkl} \frac{\partial}{\partial r_j} \left(\phi(c) \frac{\partial}{\partial r_k} \right) \right] u_l^*(r) &= C_{ijkl}^{eff} \epsilon_T \delta_{kl} \frac{\partial \beta(c)}{\partial r_j} - \Delta C_{ijkl} E_{kl} \frac{\partial \phi(c)}{\partial r_j} \\ &\quad + \Delta C_{ijkl}^{eff} \epsilon_T \delta_{kl} \frac{\partial \{\phi(c) \beta(c)\}}{\partial r_j}. \end{aligned} \quad (3.30)$$

In earlier work on the role of elastic inhomogeneity [33,19], the homogeneous strain E_{ij} was assumed to take a constant value. For example, in Ref. [33] it is assumed to be zero, while in Ref. [19], where microstructural evolution is studied under an applied stress, a constant strain is imposed on the system through a constant E_{ij} tensor. In contrast, our present formulation allows for a prescribed overall stress (which is achieved by controlling the strain). This approach is known as “stress-control based on strain-control” (see Appendix D of Ref. [31]).

3.3.2 Computational algorithm

The equation of mechanical equilibrium, Eq. (3.30), can be solved numerically using an iterative solver such as the Liebmann method. We begin with the zeroth order approximation

to the solution by assuming that $\Delta C_{ijkl} = 0$. Thus, Equation reduces to:

$$C_{ijkl}^{eff} \frac{\partial^2 u_l^*}{\partial r_j \partial r_k} = C_{ijkl}^{eff} \epsilon_T \delta_{kl} \frac{\partial \beta(c)}{\partial r_j} \quad (3.31)$$

Taking $\epsilon_{ij}^T = C_{ijkl}^{eff} \epsilon_T \delta_{kl}$, we can write the above equation as:

$$C_{ijkl}^{eff} \frac{\partial^2 u_l^*}{\partial r_j \partial r_k} = \epsilon_{ij}^T \frac{\partial \beta(c)}{\partial r_j} \quad (3.32)$$

For $i=1,2$ above equation will form as two linear equations in u_x and u_y . We can solve these equations using the Liebmann method. After solving for u_x and u_y , we can calculate the elastic strain and stress fields using Eq. (3.29) and Eq. (3.13), respectively. We can then calculate the elastic contribution to the chemical potential μ^{el} by taking the functional derivative of F^{el} with respect to composition. Finally, we can substitute μ^{el} into the Cahn-Hilliard equation to get the final form of the Cahn-Hilliard equation with elasticity, which can be solved using a finite difference method.

3.4 Simulation Algorithm

3.4.1 Spatial Discretization

We discretize the spatial domain using a uniform grid with spacing Δx and Δy in the x and y directions, respectively. The composition field $c(x, y, t)$ is approximated at grid points $(i\Delta x, j\Delta y)$ as $c_{i,j}^n$, where n denotes the time step. The Laplacian operator is discretized using central differences:

$$\nabla^2 c_{i,j} \approx \frac{c_{i+1,j} - 2c_{i,j} + c_{i-1,j}}{(\Delta x)^2} + \frac{c_{i,j+1} - 2c_{i,j} + c_{i,j-1}}{(\Delta y)^2}, \quad (3.33)$$

and similarly for other field variables. For the chemical potential, we have

$$\mu_{i,j} = h(c_{i,j}) - 2K_c \nabla^2 c_{i,j} + \mu_{i,j}^{el}, \quad (3.34)$$

where $h(c_{i,j}) = \partial f_0 / \partial c|_{c_{i,j}}$ is the local chemical potential, and $\mu_{i,j}^{el}$ is the elastic contribution at grid point (i, j) .

3.4.2 Temporal Discretization of Cahn-Hilliard Equation

The Cahn–Hilliard equation (Eq. (3.1)) is discretized in time using a semi-implicit scheme, where the nonlinear term $h(c)$ is treated explicitly and the fourth-order derivative term is treated implicitly:

$$\frac{c_{i,j}^{n+1} - c_{i,j}^n}{\Delta t} = M \nabla^2 \mu_{i,j}^*, \quad (3.35)$$

where $\mu_{i,j}^*$ is a stabilized chemical potential. To ensure numerical stability, we use:

$$\mu_{i,j}^* = h(c_{i,j}^n) - 2K_c \nabla^2 c_{i,j}^{n+1} + \mu_{i,j}^{el,n} \quad (3.36)$$

Expanding and rearranging, the discretized update formula becomes:

$$c_{i,j}^{n+1} - M \Delta t \cdot 2K_c \nabla^4 c_{i,j}^{n+1} = c_{i,j}^n + M \Delta t \nabla^2 \left(h(c_{i,j}^n) + \mu_{i,j}^{el,n} \right), \quad (3.37)$$

where $\nabla^4 = \nabla^2(\nabla^2)$ is the biharmonic operator. This is a linear system in c^{n+1} that can be solved iteratively at each time step.

3.4.3 Temporal Discretization of Allen-Cahn Equation

The Allen–Cahn equation (Eq. (3.2)) is discretized explicitly in time:

$$\frac{\eta_{i,j}^{n+1} - \eta_{i,j}^n}{\Delta t} = -L \left[\frac{\delta(F^{ch}/N_v)}{\delta \eta} \Big|_{i,j}^n + \frac{\delta(F^{el}/N_v)}{\delta \eta} \Big|_{i,j}^n \right]. \quad (3.38)$$

where $\delta F^{ch} / \delta \eta$ includes the double well and gradient terms:

$$\frac{\delta(F^{ch}/N_v)}{\delta \eta} = \frac{\partial f(c, \eta)}{\partial \eta} - 2K_\eta \nabla^2 \eta. \quad (3.39)$$

3.4.4 Mechanical Equilibrium Solver

At each time step, the mechanical equilibrium equation (Eq. (3.30)) is solved using the iterative Liebmann method. The displacement field $u_{l,i,j}^*$ is updated at each iteration using:

$$u_{l,i,j}^{*(k+1)} = u_{l,i,j}^{*(k)} + \omega \cdot \text{Residual}_{i,j}^{(k)} \quad (3.40)$$

where ω is a relaxation parameter (typically $\omega \in [0.5, 1.5]$), $\text{Residual}_{i,j}^{(k)}$ is the residual from Eq. (3.30) computed using Eq. (3.33), and k denotes the iteration count within a time step.

3.4.5 Numerical Algorithm

The overall time-stepping algorithm is:

1. Given $c_{i,j}^n$ and $\eta_{i,j}^n$, compute $h(c_{i,j}^n)$ at all grid points.
2. Assemble the composition-dependent elastic moduli and eigenstrain fields.
3. Solve the mechanical equilibrium (Eq. (3.30)) iteratively using Liebmann method until $\text{Error} < \text{tol}$ (typically 10^{-8}), obtaining $u_{i,j}^*$.
4. Compute the elastic strain and stress from the displacement field using Eq. (3.33).
5. Calculate the elastic chemical potential $\mu_{i,j}^{el}$ by functional derivative of F^{el} .
6. Solve the discretized Cahn-Hilliard equation (Eq. (3.37)) for $c_{i,j}^{n+1}$.
7. Update the order parameter using Eq. (3.38) to obtain $\eta_{i,j}^{n+1}$.
8. Advance to time step $n + 1$ and repeat.

3.4.6 Convergence and Error Control

Convergence of the mechanical equilibrium solver is monitored using the discrete L^2 -norm:

$$\text{Error}^{(k)} = \left(\sum_{i,j} (u_{i,j}^{(k+1)} - u_{i,j}^{(k)})^2 \right)^{1/2}. \quad (3.41)$$

The solution is considered converged when $\text{Error}^{(k)} < \text{tol}_{\text{mech}}$. Additionally, to enhance stability and reduce computation, from the second time step onwards the displacement solution from the previous time step is used as the initial guess for the Liebmann iterations.

Chapter 4

Results and Discussion

Begin writing your results here.

Chapter 5

Conclusion

Begin writing your conclusion here.